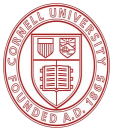


Examining *A Watermark for Large Language Models*



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Original Paper: Kirchenbauer et al. (2023), *A Watermark for Large Language Models*



Introduction

- This project investigates watermarking in language model outputs, focusing on embedding detectable signals during text generation without degrading quality

Problem/Motivation

- As large language models become widely used, it is increasingly difficult to distinguish AI-generated text from human-written content. This raises concerns around misinformation, plagiarism, and lack of accountability

Goal

- Reproduce statistical watermark detectability and perplexity results reported in the paper

Methodology

- Model: OPT-350M (the paper used OPT-6.7B)
- Data: allenai/c4 library (same as paper)

DATA CLEANING PIPELINE

1 Start with Raw Text

text1, text2, text3, ...

Raw Text Examples



text1

The quick brown fox jumps over the lazy dog.



text2

Deep learning models require large amounts of data.



text3

Natural language processing is fascinating.

⋮

2 Convert to Array of Token IDs

Tokenize each text into token_ids

Token IDs (Arrays)

[101, 1996, 4248, 2829, 4419, 102]

[2023, 2058, 1996, lazy, 3899, 1012, ...]

Length = 312

[101, 4280, learning, 2942, require, 2201, 2766, amounts, 1997, data, ...]

Length = 278

[101, 3936, language, 2537, is, 16353, 1012, ...]

Length = 145

⋮

3 Keep Only Valid Lengths

Keep only token_ids with length between 250 and 400 (inclusive)

Filtered / Cleaned Data



[101, 1996, 4248, 2829, 4419, 102]

[2023, 2058, 1996, lazy, 3899, 1012, ...]

Length = 312



[101, 4280, learning, 2942, require, 2201, 2766, amounts, 1997, data, ...]

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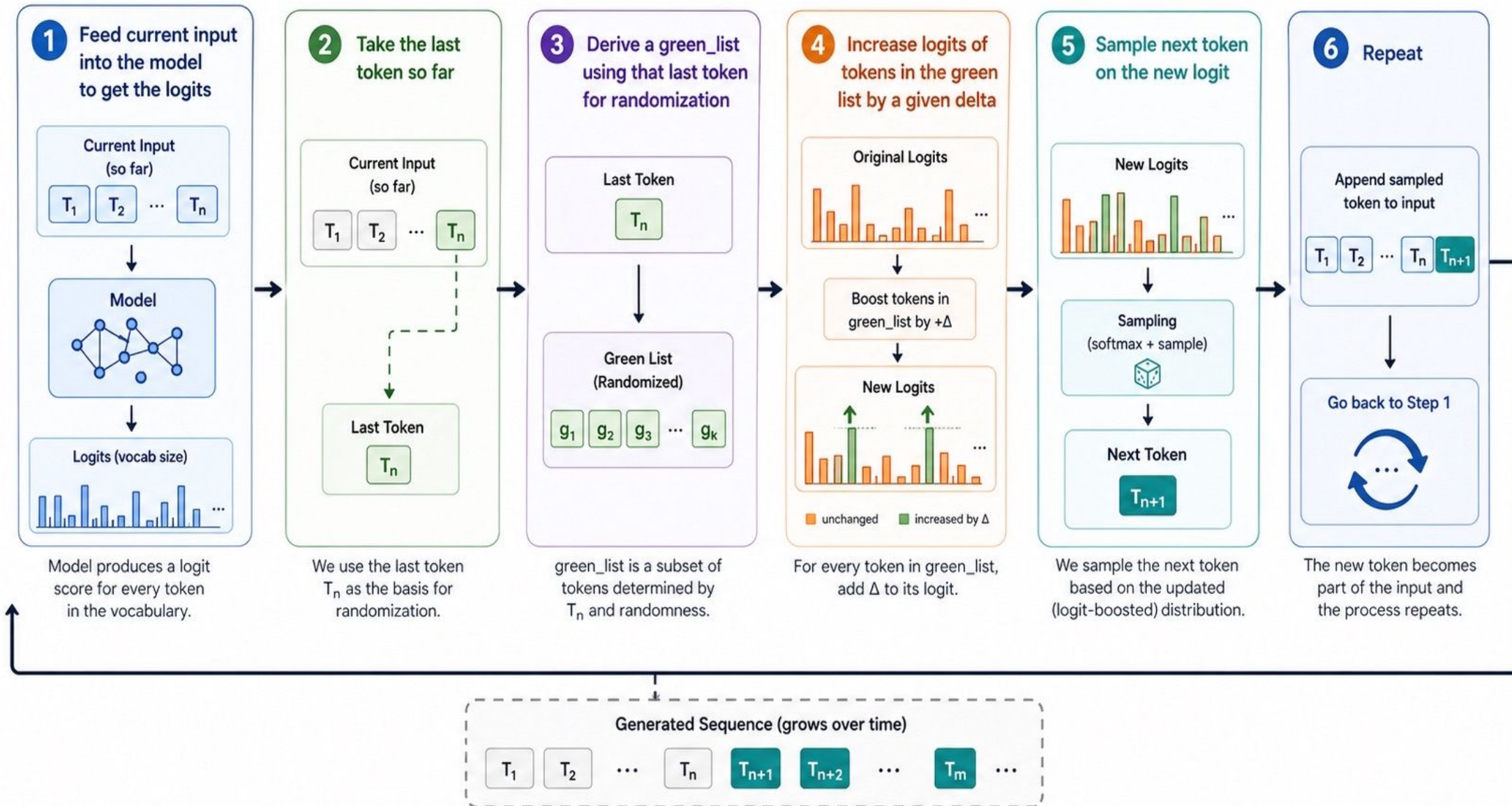
Result: Cleaned dataset containing only sequences with $250 \leq \text{length} \leq 400$



Ready for downstream tasks (model training, fine-tuning, etc.)

TEXT GENERATION PROCESS (ONE STEP)

Repeat until stop condition (e.g., EOS token or max length)



GREEN-LIST DETECTION PROCESS

Scan each token in the text and compute the z-score

Input Text (Tokenized)

Current Token



Total number of tokens = T

1 Generate green list based on the current token

Use current token T_i (and randomization) to generate the green list.



Green List _{i}

g_1 g_2 g_3 ... g_k

(subset of vocabulary)

2 Check if current token is in the green list

Is T_i in Green List _{i} ?



Yes
(1)



No
(0)

3 Increment total green count if in green list

If Yes (1):
 $G = G + 1$
else:
 G unchanged

Total Green Count
 G

4 Repeat for each token



Move to next token (T_{i+1}) and repeat Steps 1–4 until all T tokens are processed.

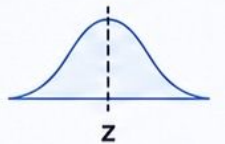
5 Plug into the z-score formula

$$z = \frac{G - y \cdot T}{\sqrt{T \cdot y \cdot (1 - y)}}$$

Where:

- G = total green count
- T = total number of tokens
- y = expected green rate (= green list size / vocab size)

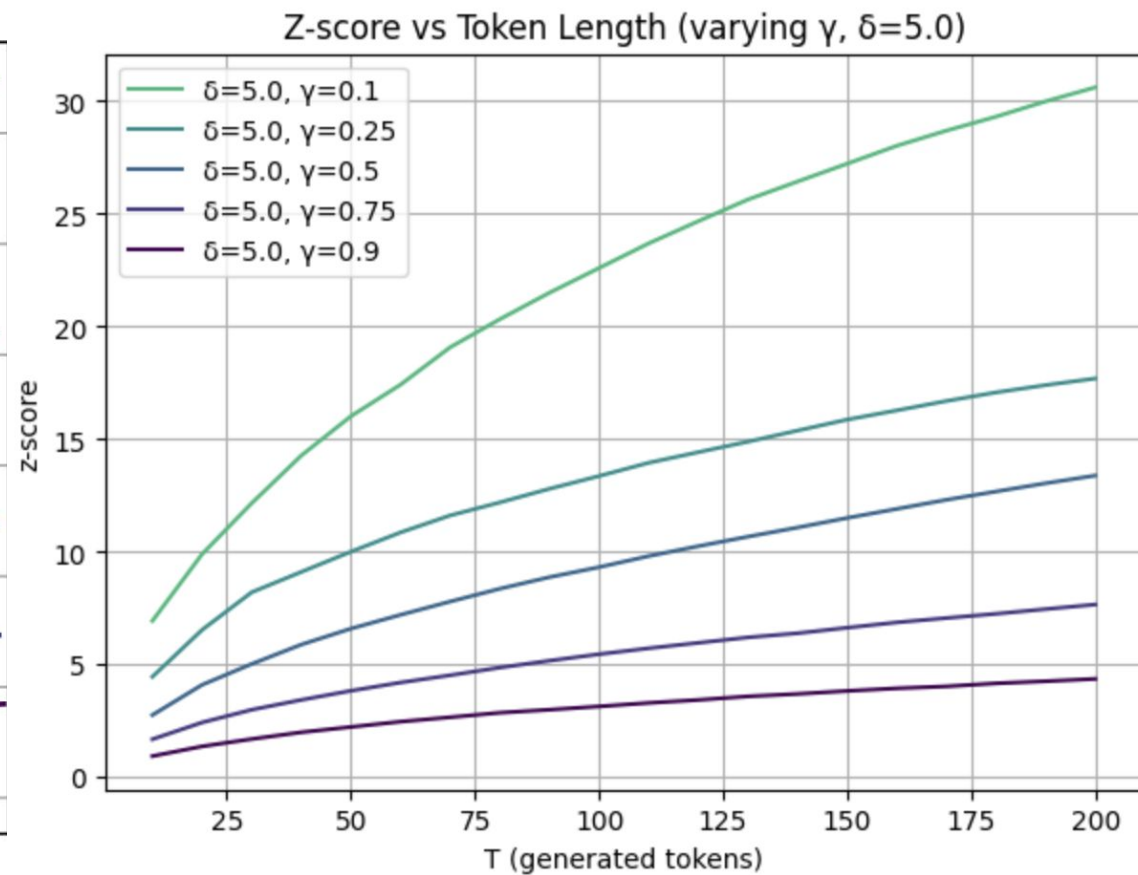
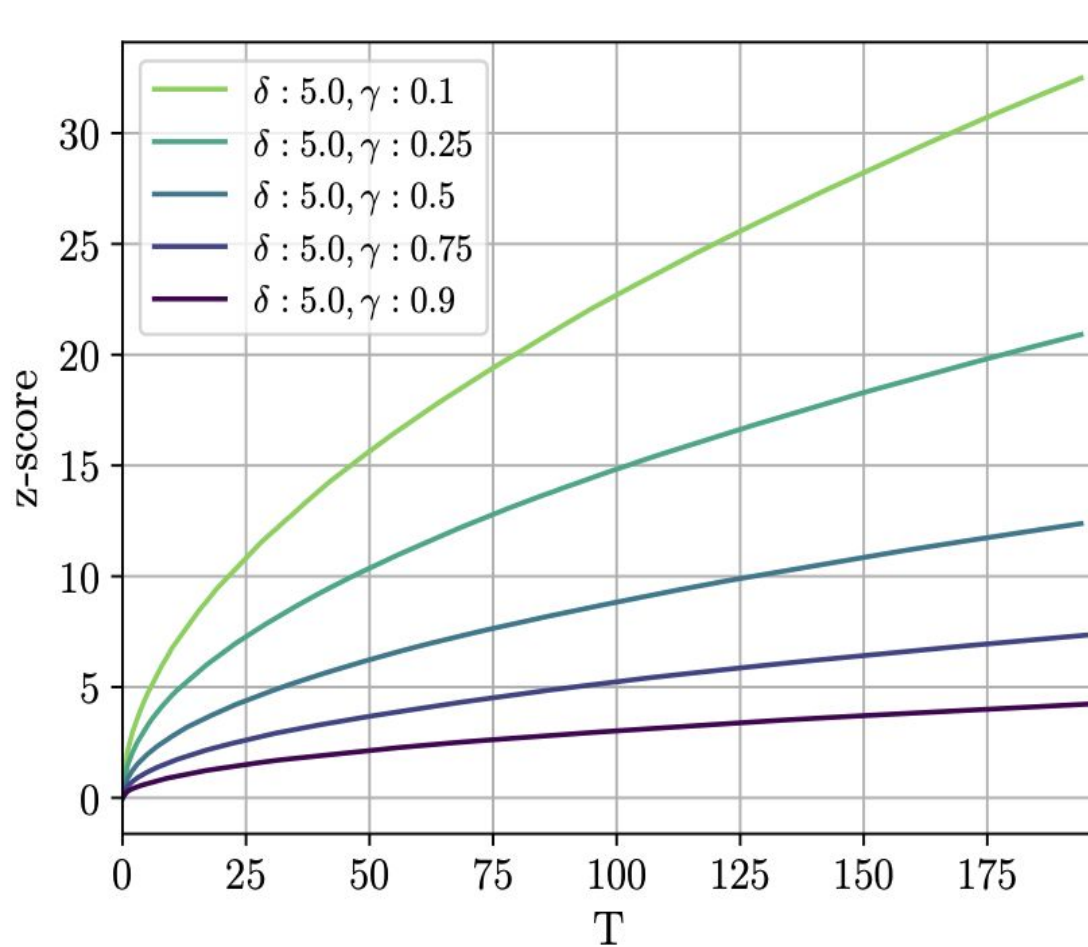
Detection Result



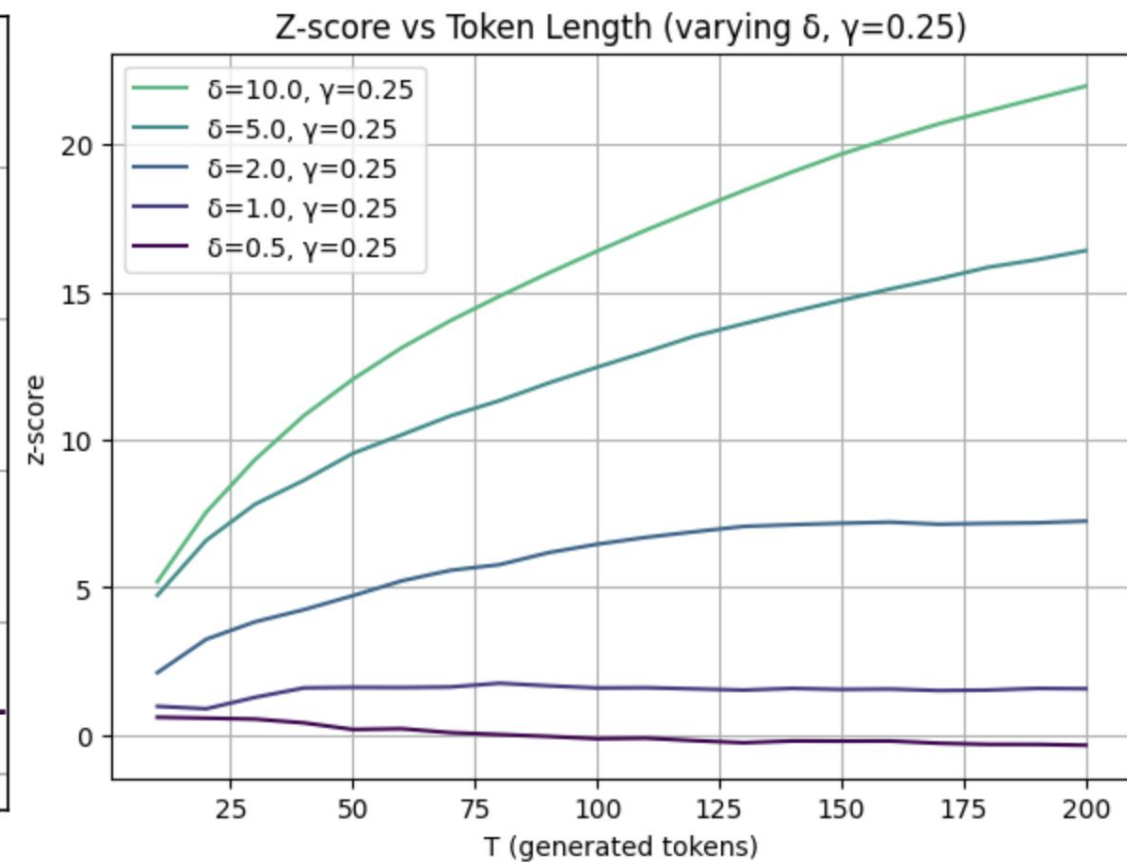
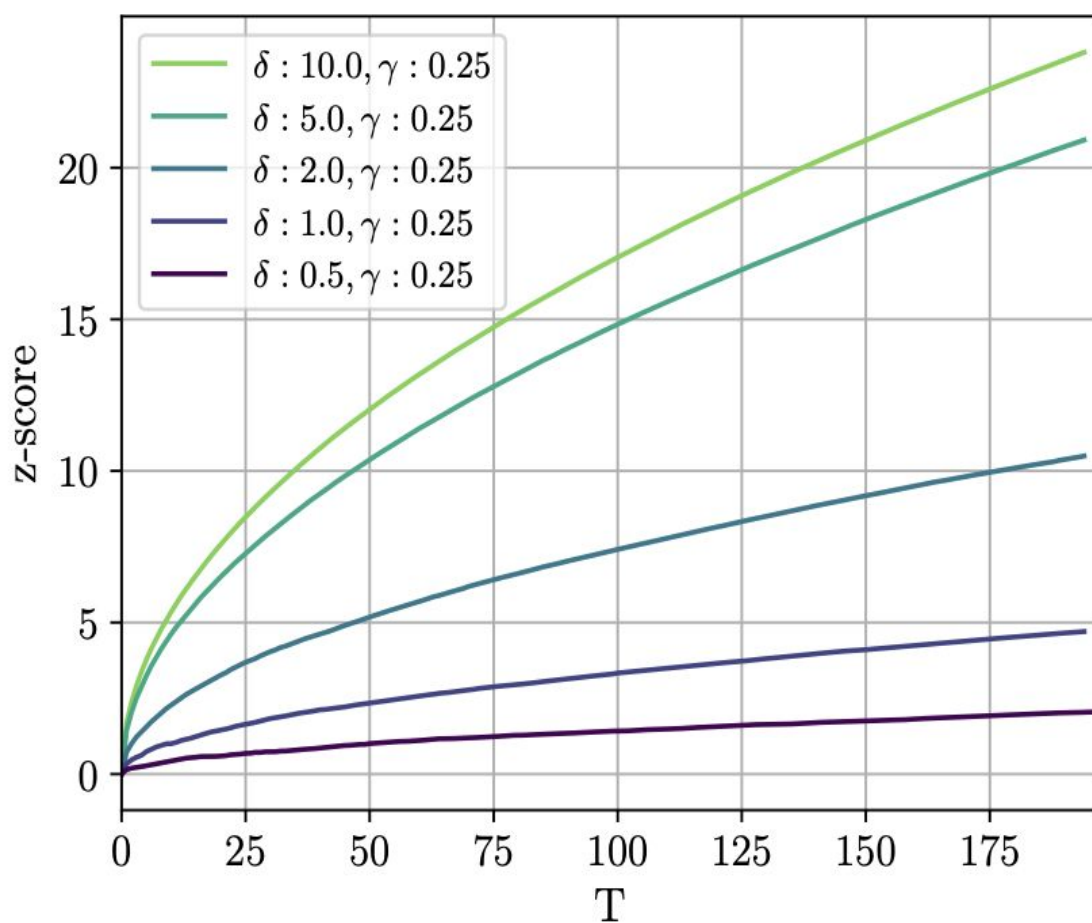
Use z-score to determine anomalous (machine-generated) vs. normal (human-written).

Loop over all tokens: $i = 1, 2, 3, \dots, T$

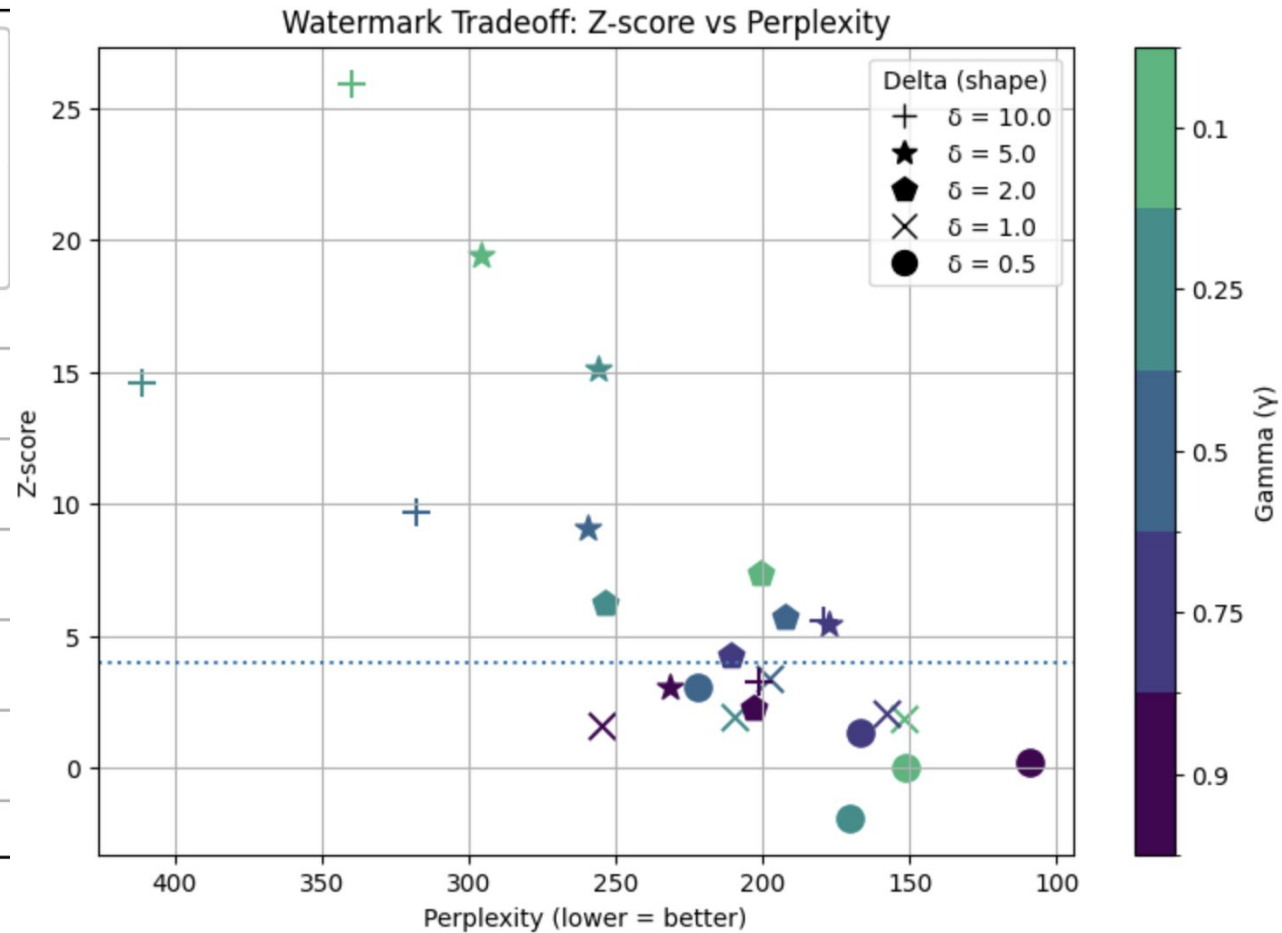
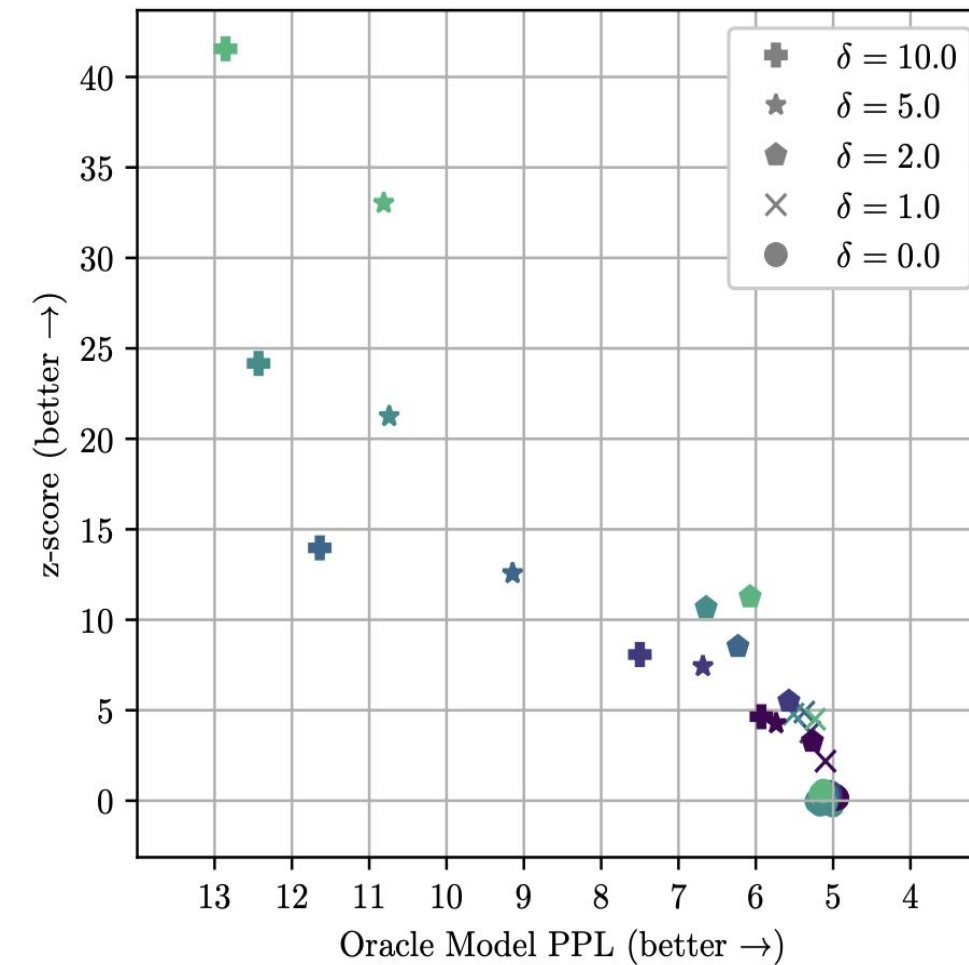
Reimplementing Graphs from the Paper



Graph from the Paper | Our Reimplementation Results



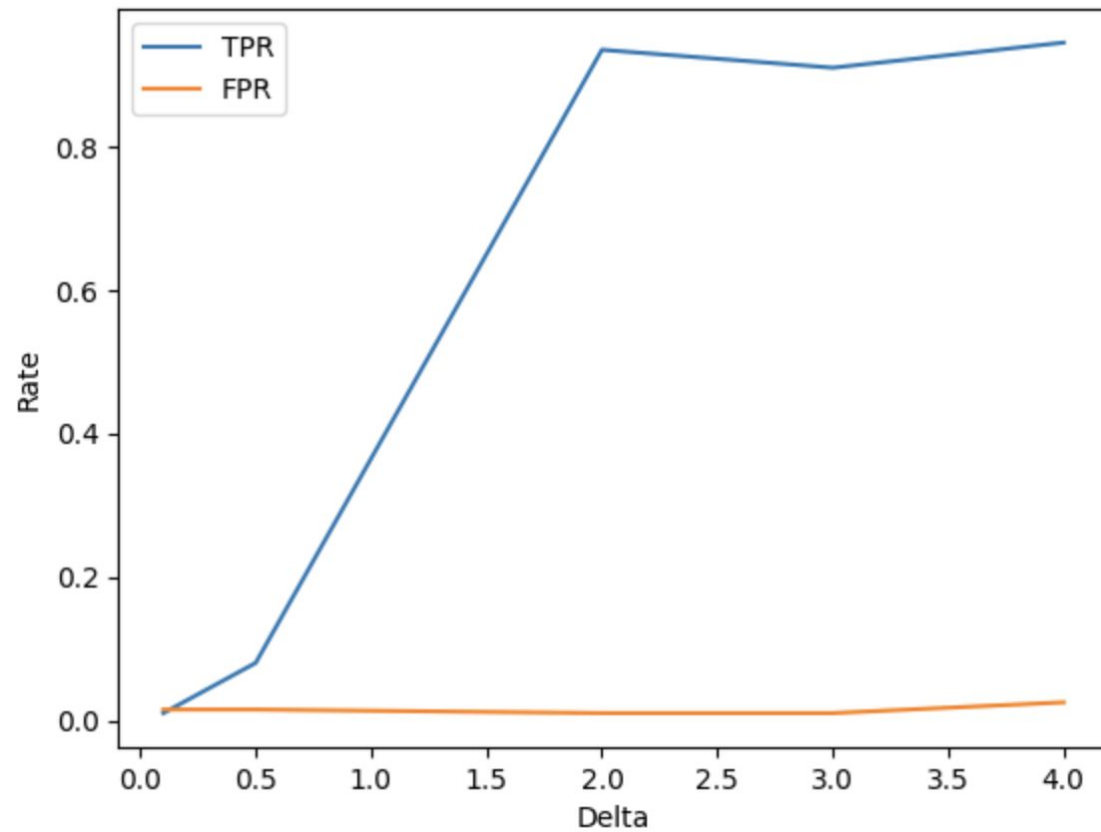
Graph from the Paper | Our Reimplementation Results



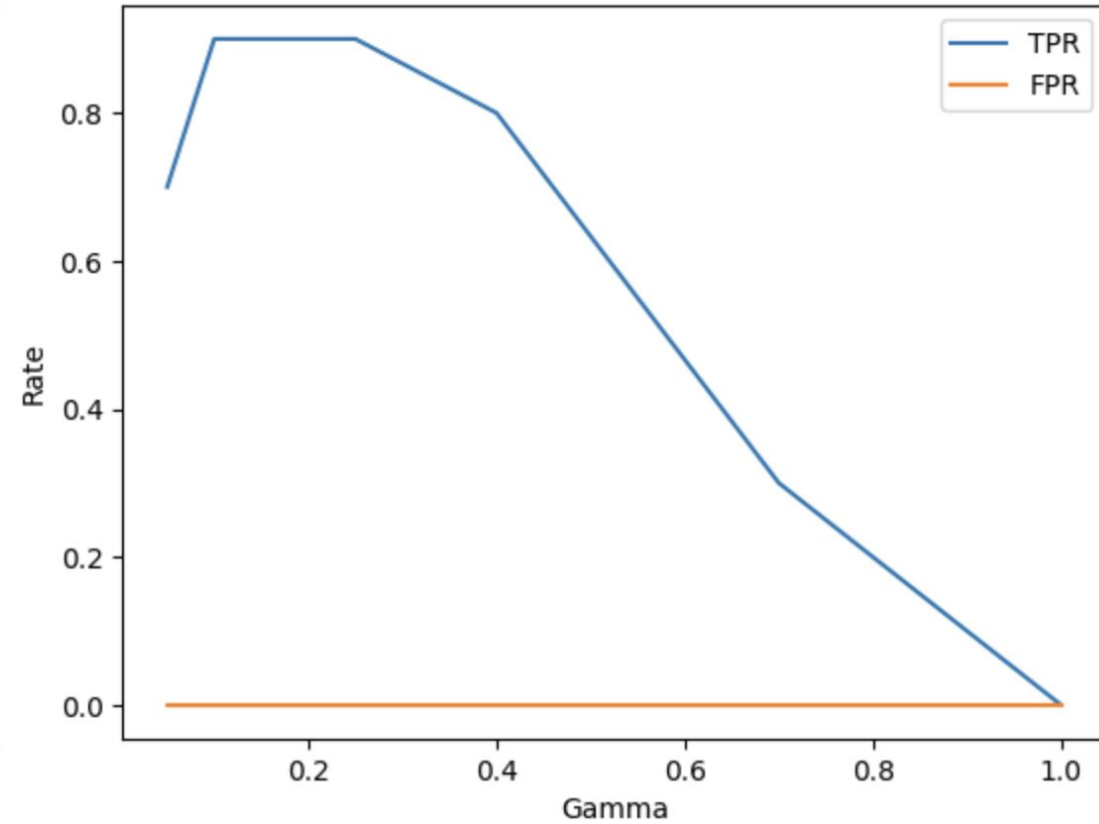
Graph from the Paper | Our Reimplementation Results

New Experiments

Effect of Delta on Watermark Detection



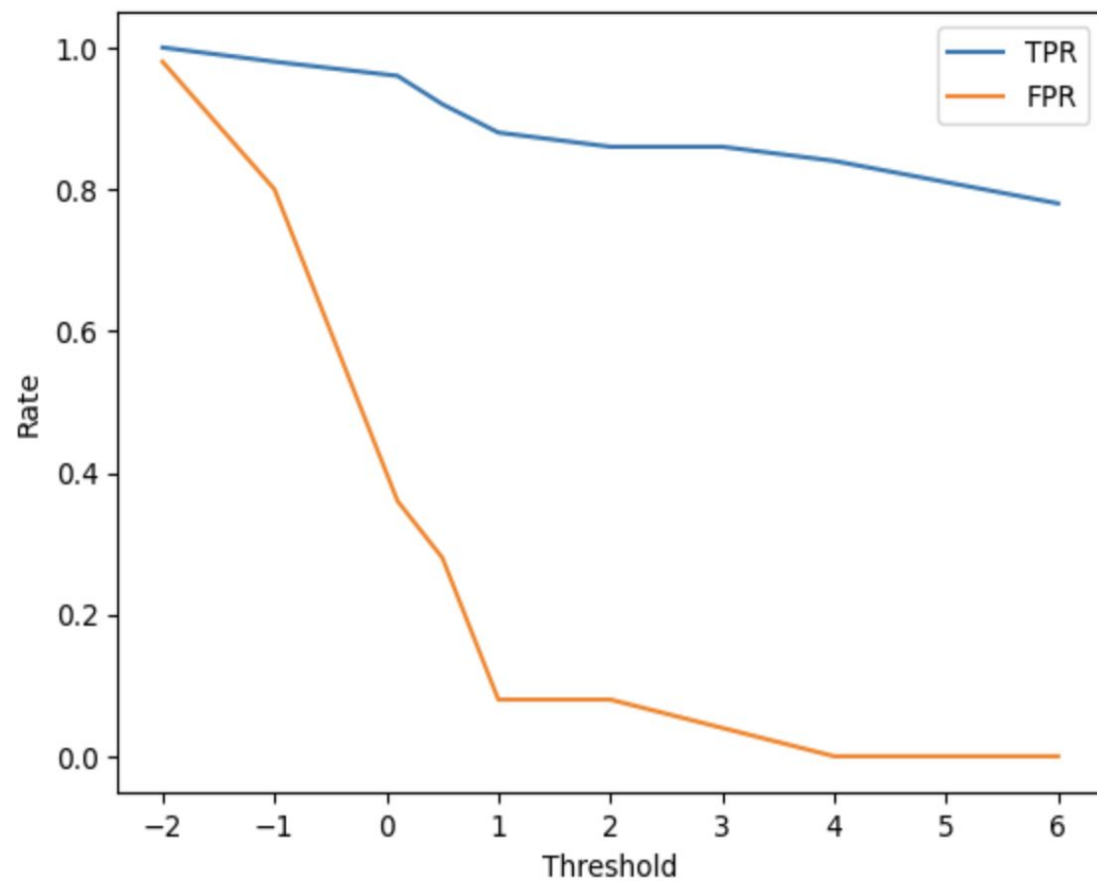
Effect of Gamma on Watermark Detection



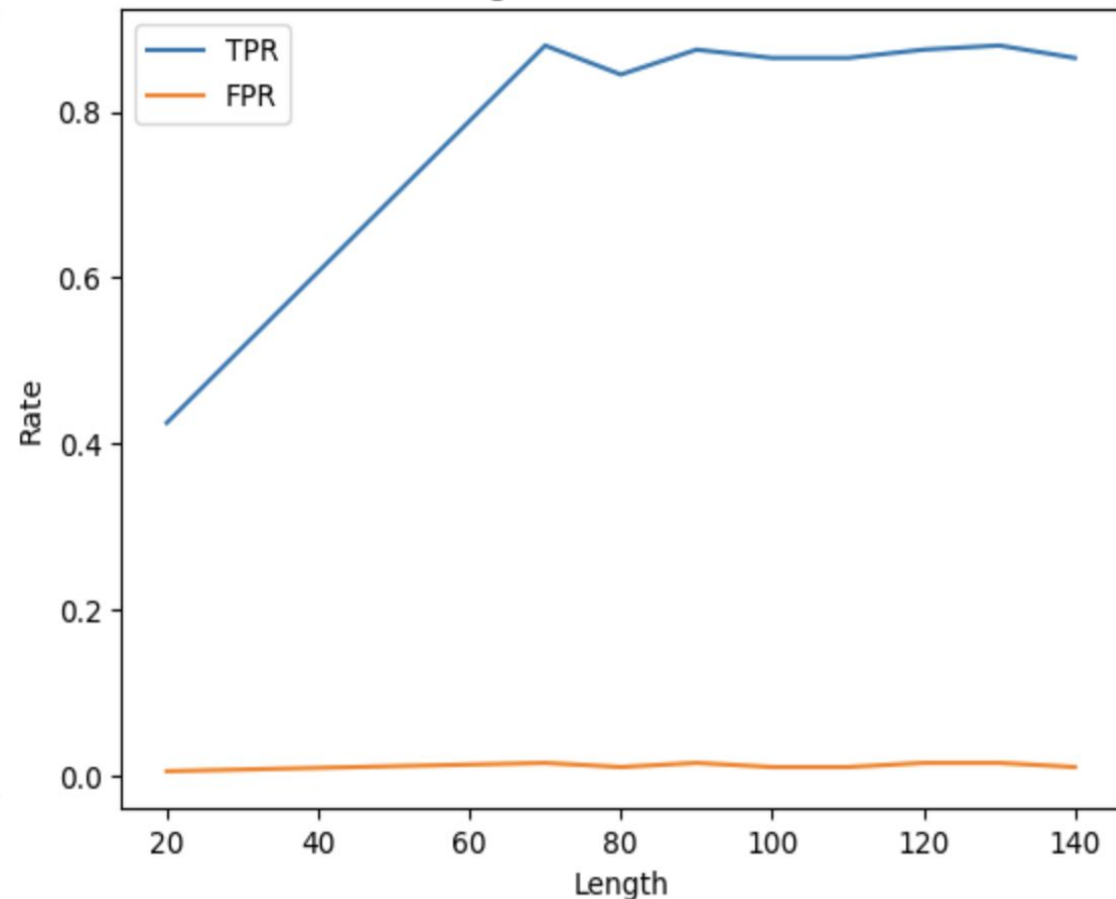
TPR = True Positive Rate | FPR = False Positive Rate

Purpose: Examine the Paper's Chosen Parameters of $\delta=2$, $\gamma=0.25$, threshold=4

Effect of Threshold on Watermark Detection



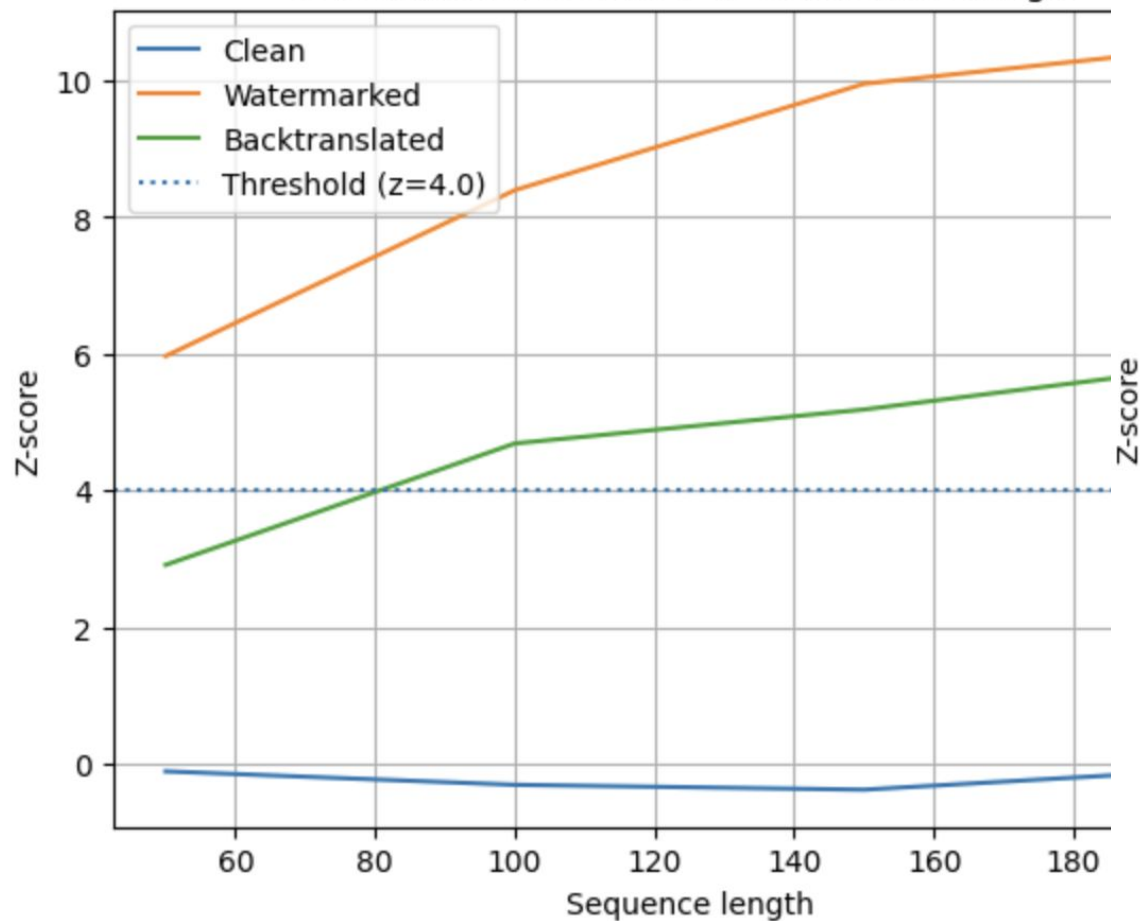
Effect of Length on Watermark Detection



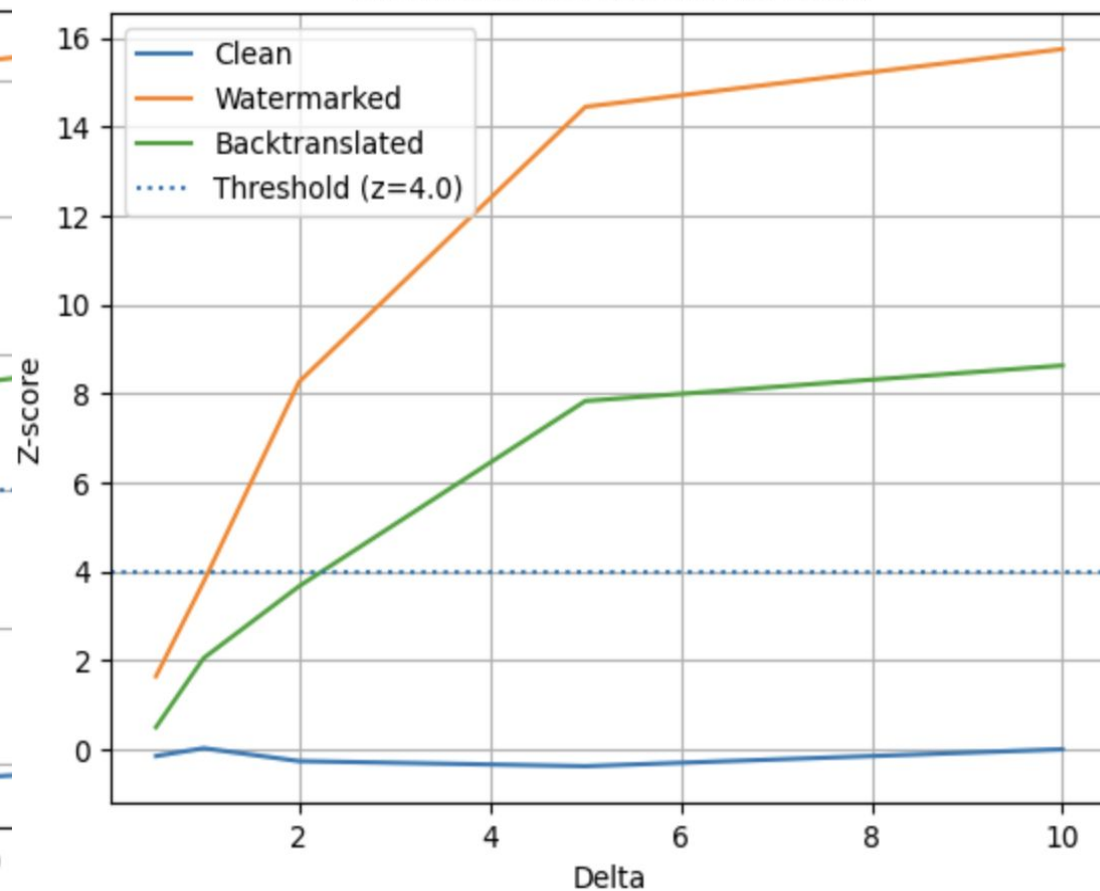
TPR = True Positive Rate | FPR = False Positive Rate

Purpose: Examine the Paper's Chosen Parameters of $\delta=2$, $\gamma=0.25$, $\text{threshold}=4$

Backtranslate: Z-score vs Generation Length



Backtranslate: Z-score vs Delta



Bactranslation Attack:

Watermarked Text English -> German -> English
Will the Watermark Survive being Translated?